

AIAC-Ass-15.3

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Batch:05

Task 1–Real-Time Applicaton: Online Food Ordering API

Scenario:

Design a simple API for an online food ordering system.

Requirements:

Endpoints required:

Endpoints required:

- GET /menu → List available dishes
- POST /order → Place a new order
- GET /order/{id} → Track order status
- PUT /order/{id} → Update an existing order (e.g., change items)
- DELETE /order/{id} → Cancel an order
- AI should generate:
- REST API code
- Suggested improvements (like authentication, pagination)

Expected Output:

- Fully working API simulating a food ordering backend.

Prompt:

Create a simple REST API for an Online Food Ordering System using Node.js and Express.

Include endpoints:

- GET /menu → list food items
- POST /order → create order
- GET /order/:id → get order status
- PUT /order/:id → update order

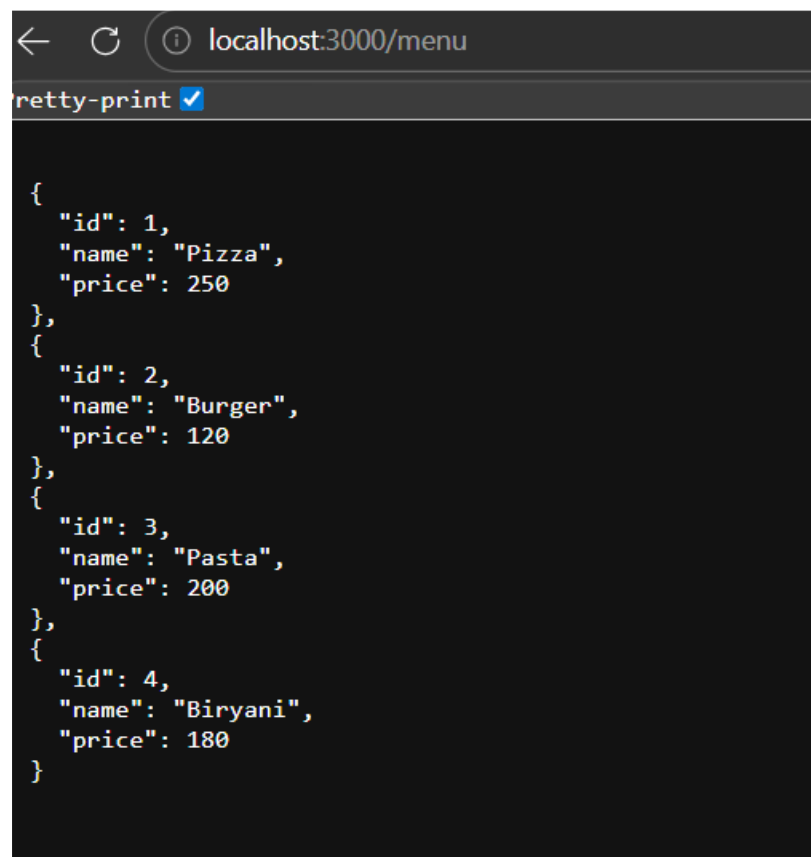
- DELETE /order/:id → cancel order

Use in-memory storage, UUID for IDs, proper status codes, and return clean JSON responses.
Provide

complete working code in one file (server.js) with run instructions. Also suggest improvements like

authentication and pagination.

Output:



A screenshot of a web browser window. The address bar shows 'localhost:3000/menu'. Below the address bar, there is a 'pretty-print' button with a checkmark. The main content area displays a JSON array of four menu items, each with an 'id', 'name', and 'price'.

```
{
  "id": 1,
  "name": "Pizza",
  "price": 250
},
{
  "id": 2,
  "name": "Burger",
  "price": 120
},
{
  "id": 3,
  "name": "Pasta",
  "price": 200
},
{
  "id": 4,
  "name": "Biryani",
  "price": 180
}
```

POST http://localhost:3000/order

Params Authorization Headers (9) Body Scripts Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON ▾

```
1 {
2   "items": ["Pizza", "Burger"]
3 }
```

Body Cookies Headers (7) Test Results

Pretty Raw Preview JSON ▾ ↻

```
1 {
2   "message": "Order placed successfully",
3   "order": {
4     "id": "ac0715c0-15d5-4395-9d69-f22a3777a317",
5     "items": [
6       "Pizza",
7       "Burger"
8     ],
9     "status": "Placed"
10  }
```

GET http://localhost:3000/order/ac0715c0-15d5-4395-9d69-f22a3777a317

Params Authorization Headers (9) Body Scripts Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON ▾

```
1 {
2   "items": ["Pizza", "Burger"]
3 }
```

Body Cookies Headers (7) Test Results

Pretty Raw Preview JSON ▾ ↻

```
1 [{"id": "ac0715c0-15d5-4395-9d69-f22a3777a317", "items": ["Pizza", "Burger"], "status": "Placed"}]
```

PUT http://localhost:3000/order/ac0715c0-15d5-4395-9d69-f22a3777a317

Params Authorization Headers (9) Body Scripts Settings

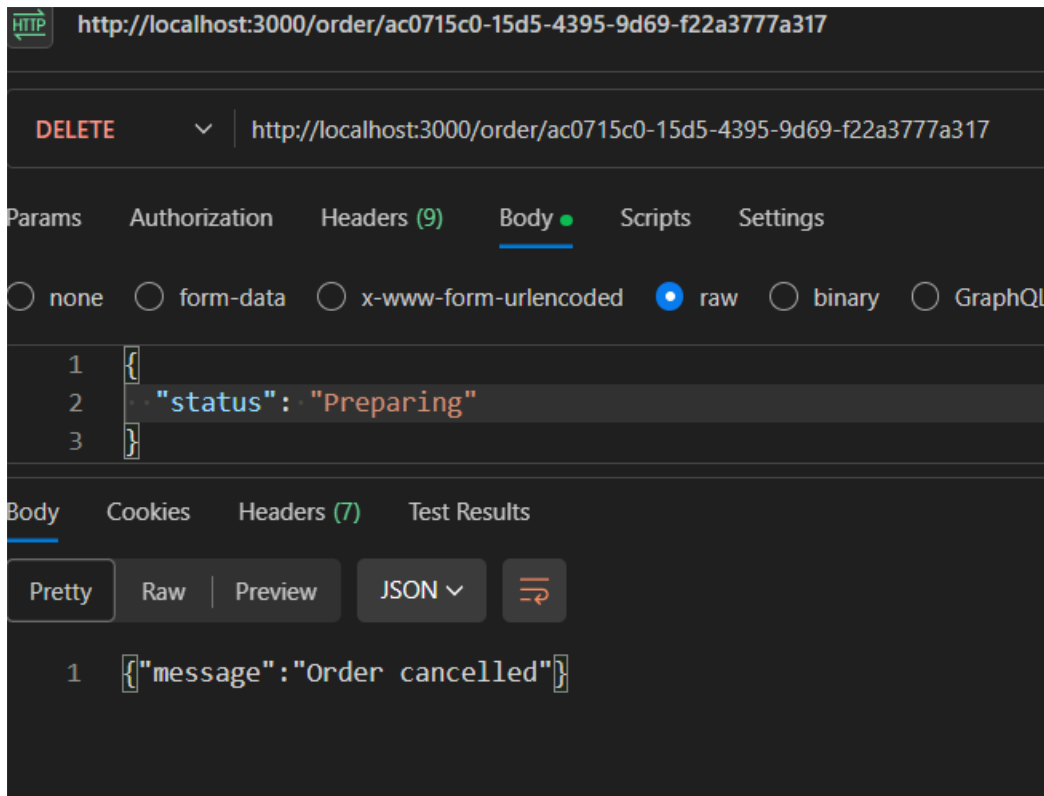
☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON ▾

```
1 {
2   "status": "Preparing"
3 }
```

Body Cookies Headers (7) Test Results Status: 200 OK Time

Pretty Raw Preview JSON ▾ ↻

```
1 [{"message": "Order updated", "order": {"id": "ac0715c0-15d5-4395-9d69-f22a3777a317", "items": ["Pizza", "Burger"], "status": "Preparing"}}]
```



Explanation : This code implements a simple API for an online food ordering system using Flask. It

allows users to view the menu, place orders, track orders, update existing orders, and cancel orders.

The menu and orders are stored in dictionaries, and the API handles various HTTP methods to

perform the necessary operations on the orders.

Task 2 – Employee Payroll API

Task:

Create an API for managing employees and their salaries.

Instructions:

- Endpoints required:
- GET /employees → List all employees
- POST /employees → Add a new employee with salary details
- PUT /employees/{id}/salary → Update salary of an employee
- DELETE /employees/{id} → Remove employee from system

- Use AI to:
- Suggest data model structure.
- Add comments/docstrings for all endpoints.

Expected Output:

- Payroll management API with validation
- Secure and structured data handling
- Clear API documentation

Prompt:

Create a REST API for an Employee Payroll System using Node.js and Express.

Include endpoints:

- GET /employees → list all employees
- POST /employees → add employee with salary details
- PUT /employees/:id/salary → update employee salary
- DELETE /employees/:id → remove employee

Use in-memory storage, unique IDs, input validation, and proper HTTP status codes. Suggest a clear

data model structure (employee id, name, role, salary). Add comments/docstrings for each endpoint.

Provide complete working code in one file and basic API documentation. Also ensure secure and

structured data handling.

Output:

HTTP <http://localhost:3000/employees>

GET <http://localhost:3000/employees>

Params Authorization Headers (9) **Body** Scripts Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON

```
1 {
2   "name": "Siri",
3   "role": "Developer",
4   "department": "IT",
5   "salary": 50000
6 }
```

Body Cookies Headers (7) Test Results

Pretty Raw Preview JSON

```
1 [
2   {
3     "id": "39aa5c0c-9fd0-48c3-a233-a0383f088b28",
4     "name": "Siri",
5     "role": "Developer",
6   }
7 ]
```

HTTP <http://localhost:3000/employees/39aa5c0c-9fd0-48c3-a233-a0383f088b28/salary>

PUT <http://localhost:3000/employees/39aa5c0c-9fd0-48c3-a233-a0383f088b28/salary>

Params Authorization Headers (9) **Body** Scripts Settings

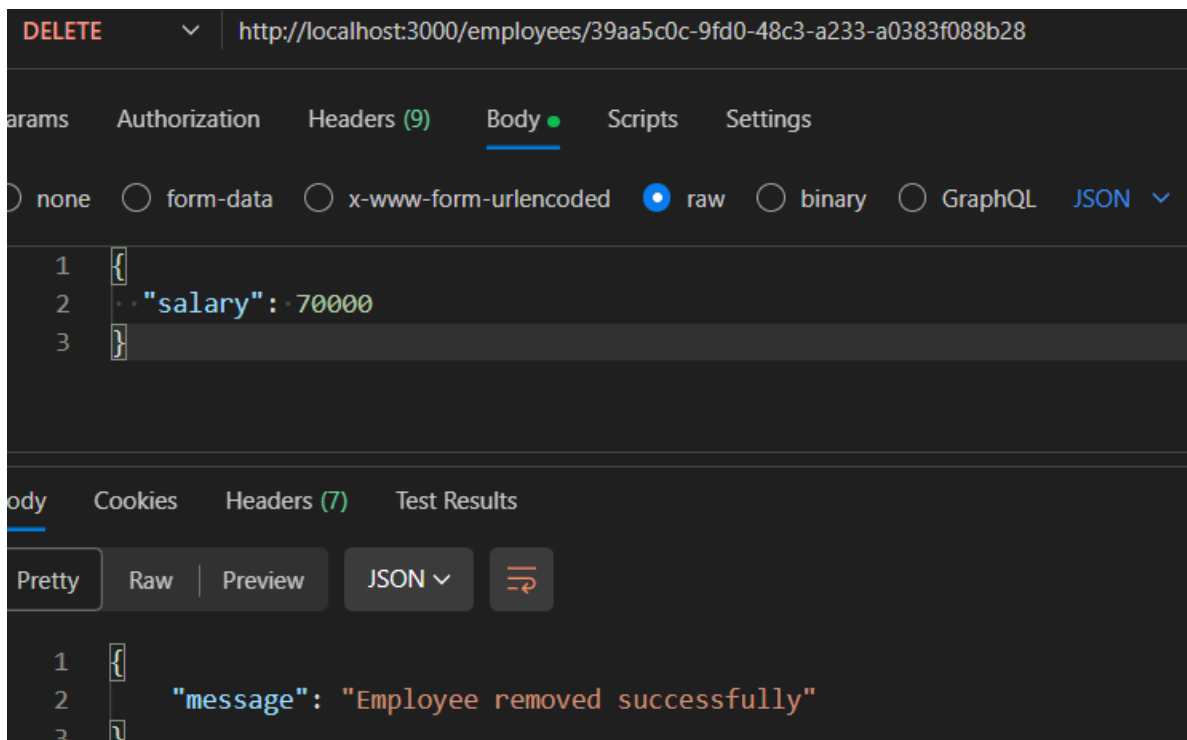
☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON

```
1 {
2   "salary": 70000
3 }
```

Body Cookies Headers (7) Test Results

Pretty Raw Preview JSON

```
1 {
2   "message": "Salary updated successfully",
3   "employee": {
4     "id": "39aa5c0c-9fd0-48c3-a233-a0383f088b28",
5     "name": "Siri",
6   }
7 }
```



Explanation : The Employee Payroll API built using Flask manages employee data through REST

endpoints.

It allows users to list employees, add new employees, update employee salary, and delete employees

using GET, POST, PUT, and DELETE methods. Employee information is stored in an in-memory list and

returned in JSON format.

Task 3: Student Information REST API

Scenario:

A university wants to digitize student records for easy retrieval. You need to build

a Student Management API that allows:

- Adding a new student (name, roll number, course, year)
- Fetching details of all students
- Updating student information
- Deleting a student record

Implementation Instructions:

1. Use Node.js + Express (or Python Flask/Django REST) with GitHub Copilot to auto-generate routes.

2. Store student data in an in-memory array or SQLite/PostgreSQL database.

Test API endpoints with Postman.

Prompt:

Create a Student Management REST API using Node.js and Express.

Features:

- Add student (name, rollNumber, course, year)
- Get all students
- Get student by ID
- Update student
- Delete student

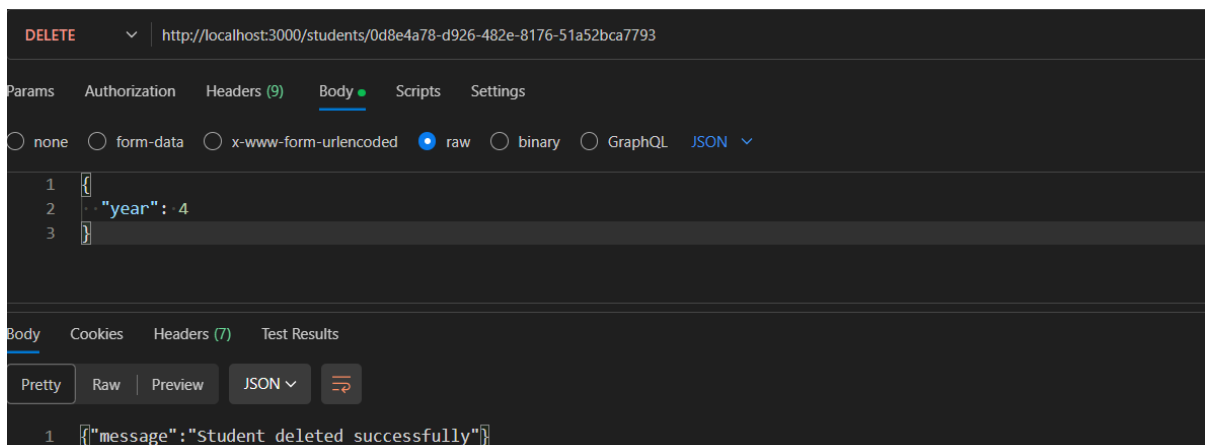
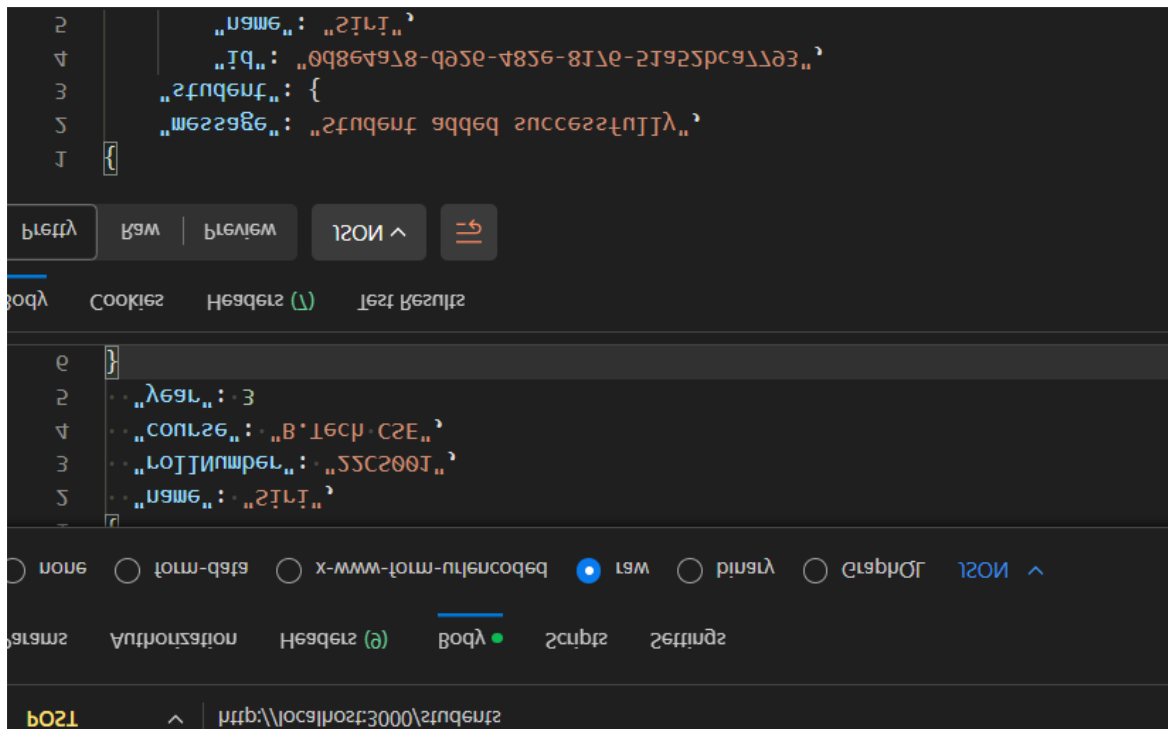
Use:

- In-memory storage
- UUID for IDs
- Input validation
- Proper status codes

Output:

- Complete server.js with comments

Output:



Explanation : This code is a simple API for a Student Information System built using the Flask framework. It allows users to perform CRUD (Create, Read, Update, Delete) operations on student

data. The API has endpoints to list all students, add a new student, update an existing student's

information, and delete a student based on their ID. The student data is stored in a list called `students`.

Task 4: Weather Information API Integration

Scenario:

Students often want to check weather before attending campus activities. Build a

backend service that fetches real-time weather data from a public API (e.g., OpenWeatherMap).

Implementation Instructions:

1. Create endpoints like:

- /weather/current/:city → fetches current weather
- /weather/forecast/:city → fetches 5-day forecast

2. Use GitHub Copilot to generate code that calls the external API.

Return results in structured JSON format with temperature, humidity, and condition.

Prompt:

Create a Weather API using Node.js and Express.

Endpoints:

- GET /weather/current/:city → current weather
- GET /weather/forecast/:city → 5-day forecast

Use OpenWeatherMap API with axios.

Return temperature, humidity, and weather condition in JSON.

Include error handling and clean code.

Output:

```
127.0.0.1:5000/weather/forecast/<city

Pretty-print ☐

{
  "city": "<city",
  "forecast": [
    {
      "condition": "Partly Cloudy",
      "day": "Monday",
      "temperature": 26
    },
    {
      "condition": "Rainy",
      "day": "Tuesday",
      "temperature": 24
    },
    {
      "condition": "Sunny",
      "day": "Wednesday",
      "temperature": 27
    }
  ]
}
```

HTTP <http://127.0.0.1:5000/weather/current/Chennai>

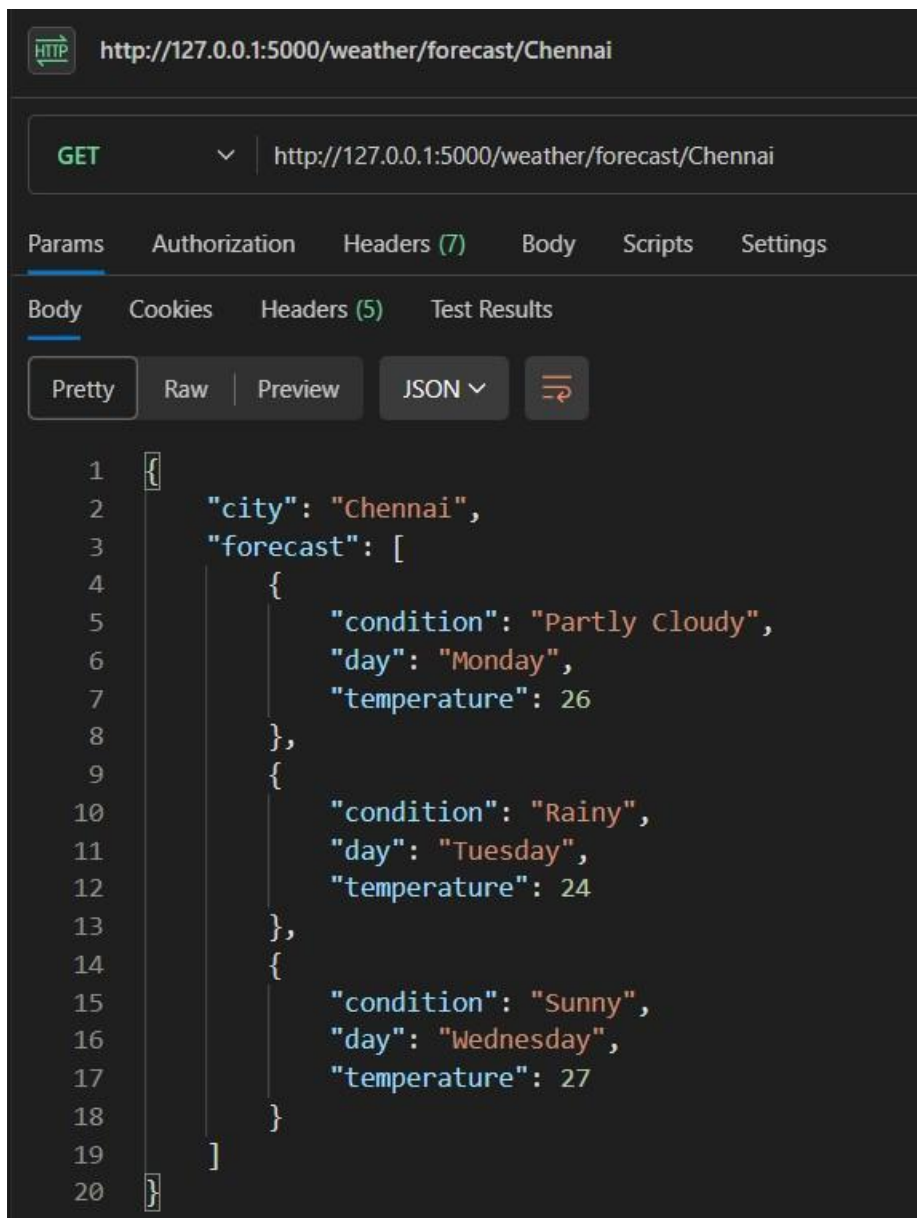
GET <http://127.0.0.1:5000/weather/current/Chennai>

Params Authorization Headers (7) Body Scripts Settings

Body Cookies Headers (5) Test Results

Pretty Raw Preview JSON

```
1 {
2   "city": "Chennai",
3   "condition": "Sunny",
4   "humidity": 60,
5   "temperature": 25
6 }
```



```
1 {
2   "city": "Chennai",
3   "forecast": [
4     {
5       "condition": "Partly Cloudy",
6       "day": "Monday",
7       "temperature": 26
8     },
9     {
10      "condition": "Rainy",
11      "day": "Tuesday",
12      "temperature": 24
13    },
14    {
15      "condition": "Sunny",
16      "day": "Wednesday",
17      "temperature": 27
18    }
19  ]
20 }
```

Explanation : This code creates a simple Flask REST API with two endpoints:

/weather/current/<city>

and /weather/forecast/<city>. The first endpoint returns mock current weather data for the specified

city, while the second endpoint returns a mock weather forecast for the next few days. In a real

implementation, you would replace the mock data with actual data fetched from the OpenWeatherMap API.

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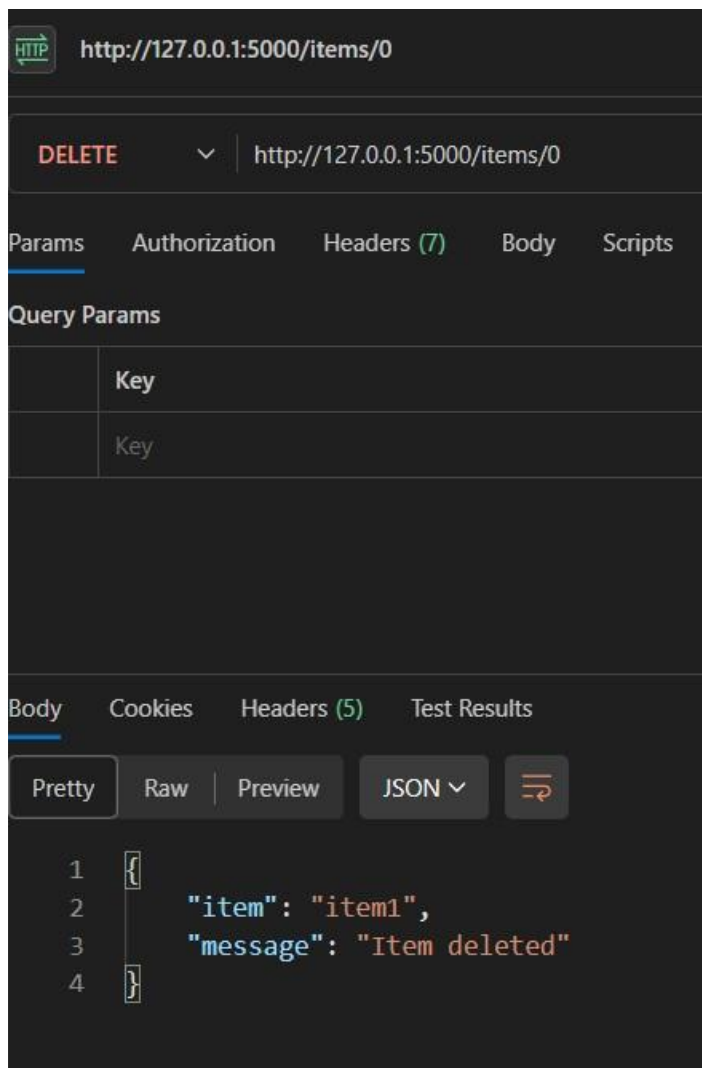
Prompt : Create a Python **Flask REST API endpoint to delete an item from a list using the DELETE

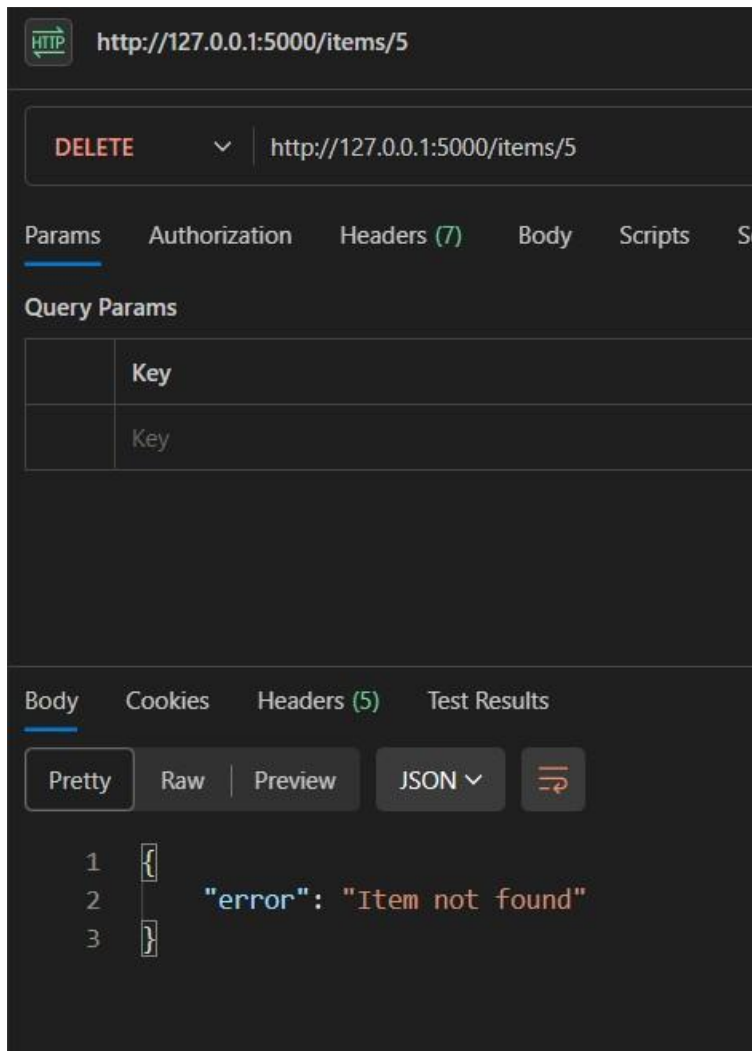
method. The endpoint should be /items/<int:index>. If the index is invalid, return a JSON error

message "Item not found" with status code 404. If successful, remove the item and return a JSON

response with "message": "Item deleted" and the deleted item.

Output:





Explanation :

This code creates a simple Flask REST API with an endpoint to delete an item from a list. The endpoint is defined as `/items/<int:index>`, where `<int:index>` is a placeholder for the index of the

item to be deleted.

When a DELETE request is made to this endpoint, the function `delete_item` is called with the index

as an argument. The function checks if the index is valid (i.e., within the bounds of the list). If it is

valid, it removes the item from the list using `pop()` and returns a JSON response indicating that the

item was deleted, along with the deleted item itself.